



Relational Conformal Prediction for Correlated Time Series

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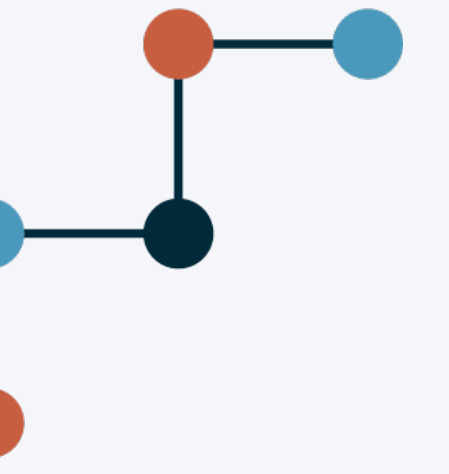
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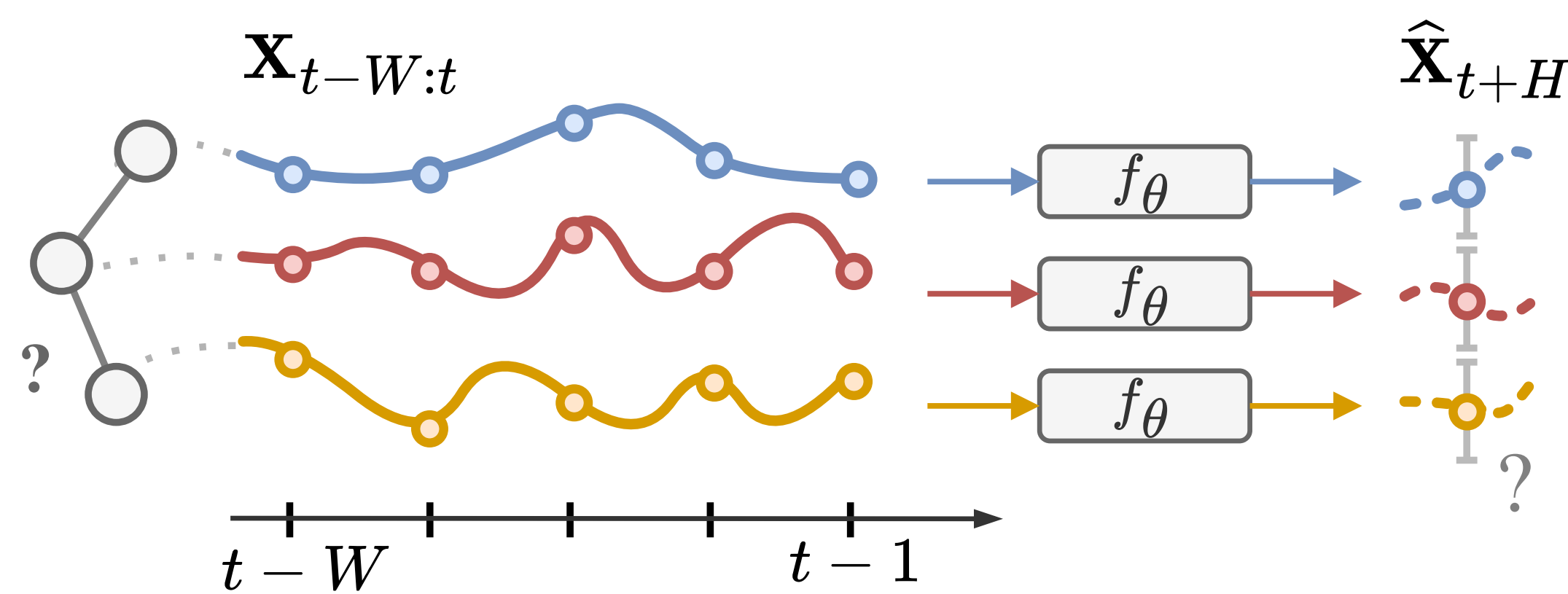


CoRel: Conformal Relational Prediction

CoRel is a novel framework for **uncertainty quantification (UQ)** in **correlated time series forecasting**.

- ⚡ Exploits **observations at related time series** to build **prediction intervals (PIs)**.
- ⚡ **Relationships** are modeled as a **graph** that can be **learned end-to-end**.
- ⚡ The resulting graph-based representation is used to learn a **GNN-based quantile predictor**.
- ⚡ Adaptive learnable **node embeddings** are used to **localize uncertainty estimates** and mitigate **non-stationarity**.

Uncertainty Quantification & Correlated Time Series Forecasting



We want to estimate a **prediction interval (PI)**, $C_{i,t}^\alpha$, for each i -th time series such that:

$$P(x_{t+H}^i \in C_{i,t}^\alpha(\hat{X}_{t+H})) \geq 1 - \alpha$$

- ! PIs should provide **coverage** while being as **narrow** as possible.
- ? Can we exploit **observations at correlated time series**?
- ? Can we use **conformal prediction** to estimate PIs in this context?

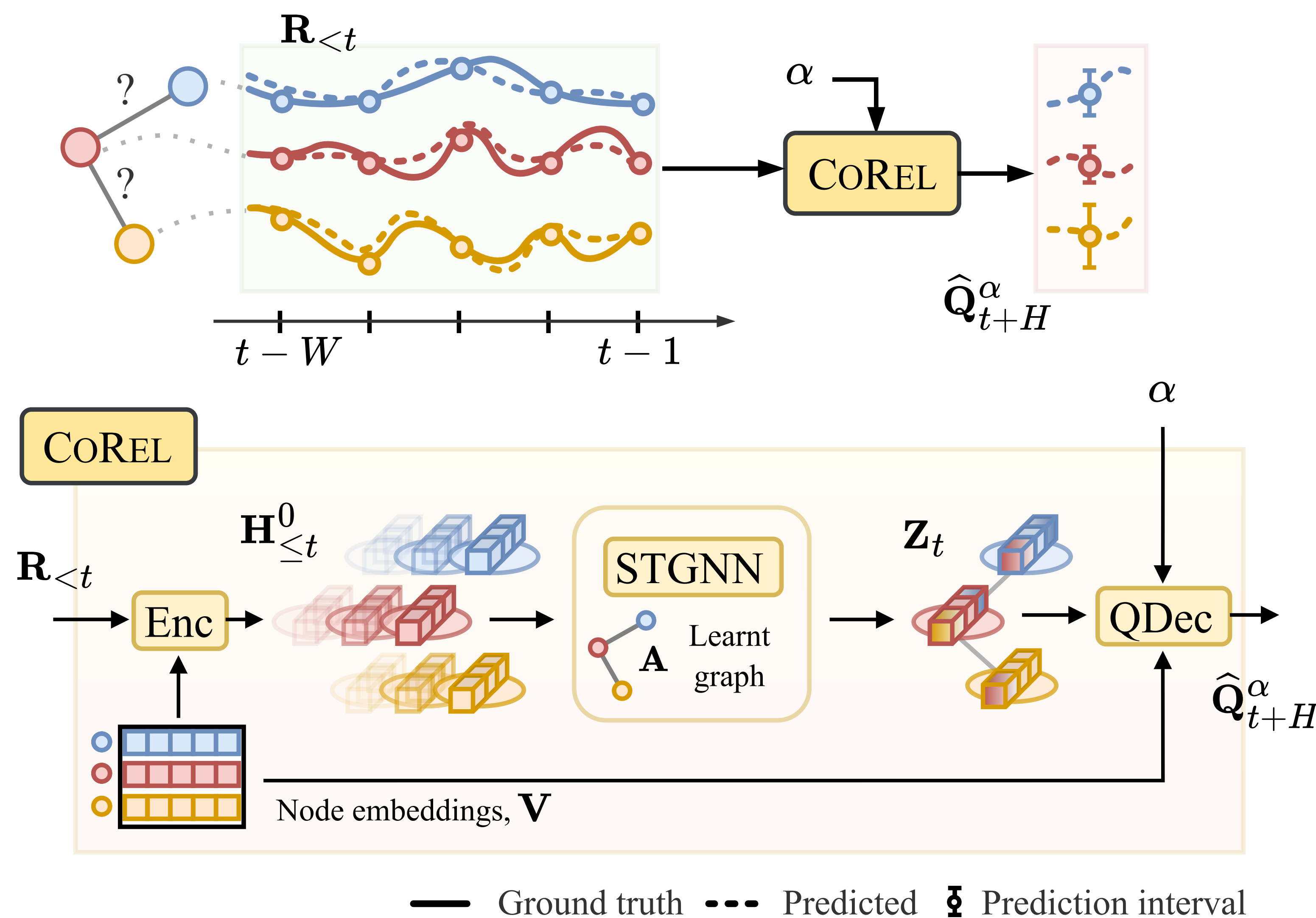
Conformal Prediction

Conformal prediction (CP) methods estimate PIs by computing **empirical quantiles** of observed **conformal scores**.

- We can use **prediction residuals** r_t^i as conformal scores
- $r_t^i = x_t^i - \hat{x}_t^i, \quad (R_t \doteq [r_t^i, r_t^{i+1}, \dots, r_t^N]^\top \in \mathbb{R}^{N \times 1}).$
- 💡 **Learn a graph** to model (**sparse**) dependencies.
- 😊 We can use **GNNs** to approximate the quantile function!
- ! Data might be **non-stationary**
→ need to be careful with **exchangeability assumptions**.

TL;DR

We leverage **graph-based operators** to model dependencies among **prediction residuals** in **conformal prediction** for **correlated time series forecasting**.



Relational Quantile Network

With relationships encoded in $A \in \mathbb{R}^{N \times N}$, we approximate the residuals' quantile function with a **spatiotemporal graph neural network (STGNN)** as

$$h_t^{i,0} = \text{ENC}(r_{t-1}^i, v^i),$$

$$Z_t = \text{STGNN}(H_{\leq t}^0, A),$$

$$\hat{q}_{t+H}^{i,\alpha} = \text{QDEC}(\alpha, z_t^i, v^i).$$

! **Node embeddings** $v^i \in \mathbb{R}^{d_v}$ are learnable parameters specific to each i -th time series.

PIs can then be computed as

$$\widehat{C}_{i,t}^\alpha = [\hat{x}_{t+H}^i + \hat{q}_t^{i,\alpha/2}, \hat{x}_{t+H}^i + \hat{q}_t^{i,1-\alpha/2}],$$

Graph Structure Learning

We learn the graph end-to-end with a **probabilistic model**.

- We learn a distribution over **K-NN graphs** with a **differentiable subset sampler**.
- For each i -th node, we sample **without replacement** K neighbors from a **categorical distribution** M_i as

$$\Phi = \mathcal{E}_\xi(R_{<t}, V, \dots)$$

$$M_i = \text{Categorical}\left(\frac{\exp\{\phi^{ik}\}}{\sum_{j=1}^N \exp\{\phi^{ij}\}}; k \in \{1, \dots, N\}\right),$$

with the **reparametrization** introduced by Xie et al., 2019¹

¹ Xie et al., "Reparameterizable Subset Sampling via Continuous Relaxations", IJCAI 2019

Adaptive Embeddings

CP methods rely on the **exchangeability** of calibration and test data.

- ☹️ Difficult to guarantee if time series are **non-stationary**.
- ! Updating the full model is **expensive**.
- 💡 We can simply **update embeddings** $V \in \mathbb{R}^{N \times d_v}$ over time!
- 😊 **Relatively small sample complexity**.

Results

Experiments on benchmark datasets

	Metric	SCP	NexCP	SPCI	HopCPT	CoRel
METR-LA	ΔCov	-1.18	-0.97	-1.13 ± 0.00	-1.22 ± 0.56	-1.02 ± 0.63
	PI-Width	20.81	25.30	19.86 ± 0.01	16.66 ± 0.13	14.17 ± 0.43
	Winkler	40.55	41.64	37.74 ± 0.01	25.37 ± 0.33	23.65 ± 0.19
AQI	ΔCov	5.07	-0.66	1.60 ± 0.01	-2.34 ± 1.19	-1.81 ± 1.67
	PI-Width	118.50	81.15	104.11 ± 0.01	80.37 ± 2.09	72.25 ± 3.16
	Winkler	150.33	132.08	145.14 ± 0.02	129.85 ± 3.14	108.71 ± 1.38

Adaptation

CER-E

	ΔCov	PI-Width	Winkler
W/o adapt.	-3.08 ± 0.15	1.84 ± 0.02	3.69 ± 0.04
W/ adapt.	-2.42 ± 0.20	1.85 ± 0.01	3.49 ± 0.04

Code available at:

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